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course/Program:- BTech CSE SY

Subject : DLM

DLM lab practical, 6

* Aim:- Verify Binary to Gray and Gray to Binary conversion using NAND gates only.

* Theory:-

i) Binary to Gray Conversion:

- i). The most significant Bit (MSB) of the gray code is always equal to the MSB of the given binary code.
- ii). other bits of the output gray code can be obtained by EX-ORing binary code bit at that index and previous index.

There are four inputs and four outputs. The input variables are defined as B_3, B_2, B_1, B_0 and the output variables are defined as G_3, G_2, G_1, G_0 from the truth table, combinational circuit is designed. The logical expressions are defined as:

$$B_3 = G_3$$

$$B_2 \oplus B_3 = G_2$$

$$B_1 \oplus B_2 = G_1$$

$$B_0 \oplus B_1 = G_0$$

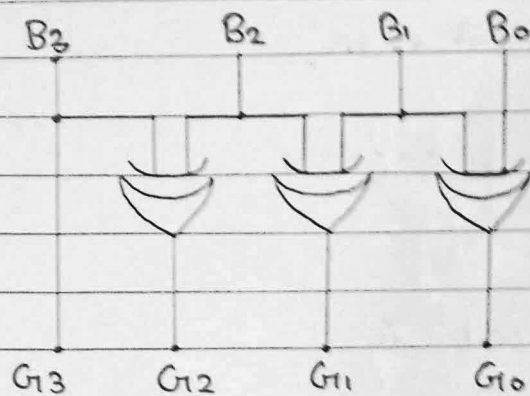


Fig.: Binary to Gray Code Converter Circuit.

Natural-binary code				Gray code			
B ₃	B ₂	B ₁	B ₀	G ₃	G ₂	G ₁	G ₀
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Fig:- Binary to Gray Code Converter Truth Table.



2) Gray to Binary.

- i) The most Significant Bit (MSB) of the binary code is always equal to the MSB of the given binary number.
- ii) other bits of the output binary code can be obtained by checking gray code bit at that Index. If current gray code bit is 0, then copy previous binary code bit, else copy invert of previous binary code bit.

There are four inputs and four outputs. The input variable are defined as B_3, B_2, B_1, B_0 . From the truth table, combinational circuit is designed. The logical expressions are defined as:

$$G_0 \oplus G_1 \oplus G_2 \oplus G_3 = B_0$$

$$G_1 \oplus G_2 \oplus G_3 = B_1$$

$$G_2 \oplus G_3 = B_2$$

$$G_3 = B_3$$

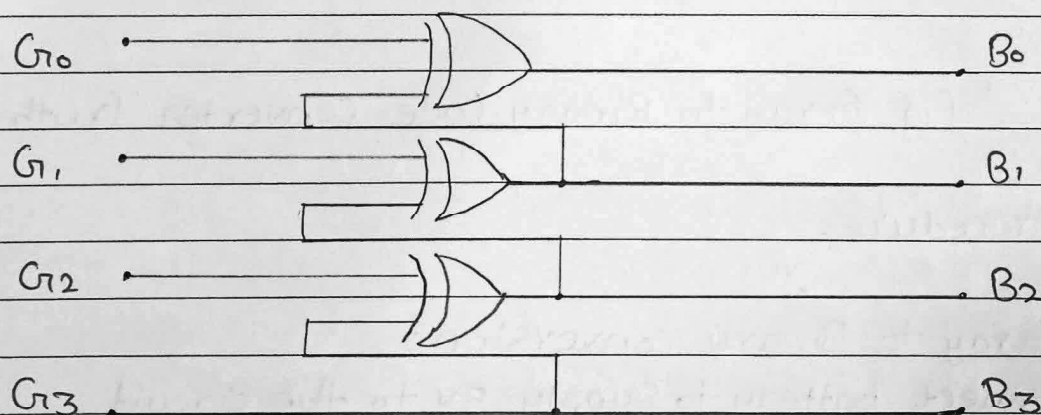


Fig: Gray to Binary code converter circuit.



Gray Code				Natural-binary code			
G ₃	G ₂	G ₁	G ₀	B ₃	B ₂	B ₁	B ₀
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0

Fig: Gray to Binary Code Converter Truth Table.

* Procedure:

① Gray to Binary conversion:

Step:1) Connect battery to supply 5V to the circuit.

Step:2) Press switches for different inputs.

Step:3) The corresponding combination of input and output LEDs lit up for different combination of inputs.



The input gray code LEDs are G_3, G_2, G_1 , and G_0 and the output binary code LEDs B_3, B_2, B_1 and B_0 glow accordingly.

Step: 5) click "Print" to get print out of the truth table.

② Binary to Gray Code conversion

Step: 1) connect battery to supply 5V to the circuit

Step: 2) Press switches for different inputs.

Step: 3) The corresponding combination of input & output LEDs lit up for different combinations of inputs.

The input binary code LEDs are B_3, B_2, B_1 and B_0 and the output gray code LEDs G_3, G_2, G_1 , and G_0 glow accordingly

Step: 4) Click "Add" to add the values to the truth table.

Step: 5) click "Print" to get the print out of the truth table.

* Conclusion:

The experiment verified binary to Gray and Gray to Binary conversions using only NAND gates, with results confirming the accuracy of the conversions.